
LECTURE NOTES

Discrete Mathematics notes

Matrices

DM549 - Discrete Mathematics

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1 Operations on matrices

Definition: Matrix Addition

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ both be $m \times n$ matrices.

The sum of A and B , denoted as $A + B$ is another $m \times n$ matrix that has $a_{ij} + b_{ij}$ as its (i, j) th entry. In other words this means that.

$$A + B = [a_{ij} + b_{ij}] \quad \forall i, j$$

Definition: Matrix Multiplication

Let A be an $m \times n$ matrix and B be a $n \times p$ matrix.

Then AB is performed as following

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix} \begin{bmatrix} b_1 & b_2 & b_3 \\ b_4 & b_5 & b_6 \\ b_7 & b_8 & b_9 \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & c_3 \\ c_4 & c_5 & c_6 \\ c_7 & c_8 & c_9 \end{bmatrix}$$

Figure 1: Matrix multiplication definition visualized

Example: Matrix Multiplication

Given two matrices A and B

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$$

Use the definition to compute AB

$$AB = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \cdot \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 30 & 24 & 18 \\ 84 & 69 & 54 \\ 138 & 114 & 90 \end{bmatrix}$$

Definition: Matrix Transpose

The transpose of a matrix is obtained by interchanging its rows and columns

$$A^T \stackrel{\text{def}}{=} \text{rows} \leftrightarrow \text{columns}$$

Given that $A = A^T$, A is called *symmetric*.

Example: Matrix Transpose

Given Matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \rightarrow A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Definition: Binary AND (meet) on matrices

The $\wedge$ (AND) operation on binary matrices are defined as individual $\wedge$ operations on each of the two matrices entries

This is also called meet, as in A meets B

$$A \wedge B = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \wedge \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} = \begin{bmatrix} a_{11} \wedge b_{11} & a_{12} \wedge b_{12} & a_{13} \wedge b_{13} \\ a_{21} \wedge b_{21} & a_{22} \wedge b_{22} & a_{23} \wedge b_{23} \\ a_{31} \wedge b_{31} & a_{32} \wedge b_{32} & a_{33} \wedge b_{33} \end{bmatrix}$$

Definition: Binary OR (join) on matrices

The $\vee$ (OR) operation on binary matrices are defined as individual $\vee$ operations on each of the two matrices entries

This is also called join, as in A joins B

$$A \vee B = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \vee \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} = \begin{bmatrix} a_{11} \vee b_{11} & a_{12} \vee b_{12} & a_{13} \vee b_{13} \\ a_{21} \vee b_{21} & a_{22} \vee b_{22} & a_{23} \vee b_{23} \\ a_{31} \vee b_{31} & a_{32} \vee b_{32} & a_{33} \vee b_{33} \end{bmatrix}$$

Definition: Boolean product on matrices

$A \odot B$ or $A \circ B$ is calculated like standard matrix multiplication, but ordinary multiplication and addition are replaced by logical operations: logical AND ($\wedge$) for multiplication and logical OR ($\vee$) for addition.

$$A \odot B = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \odot \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$$
$$= \begin{bmatrix} (a_{11} \wedge b_{11}) \vee (a_{12} \wedge b_{21}) \vee (a_{13} \wedge b_{31}) & (a_{11} \wedge b_{12}) \vee (a_{12} \wedge b_{22}) \vee (a_{13} \wedge b_{32}) & (a_{11} \wedge b_{13}) \vee (a_{12} \wedge b_{23}) \vee (a_{13} \wedge b_{33}) \\ (a_{21} \wedge b_{11}) \vee (a_{22} \wedge b_{21}) \vee (a_{23} \wedge b_{31}) & (a_{21} \wedge b_{12}) \vee (a_{22} \wedge b_{22}) \vee (a_{23} \wedge b_{32}) & (a_{21} \wedge b_{13}) \vee (a_{22} \wedge b_{23}) \vee (a_{23} \wedge b_{33}) \\ (a_{31} \wedge b_{11}) \vee (a_{32} \wedge b_{21}) \vee (a_{33} \wedge b_{31}) & (a_{31} \wedge b_{12}) \vee (a_{32} \wedge b_{22}) \vee (a_{33} \wedge b_{32}) & (a_{31} \wedge b_{13}) \vee (a_{32} \wedge b_{23}) \vee (a_{33} \wedge b_{33}) \end{bmatrix}$$

2 Concepts

Definition: Matrix

A matrix A is a rectangular grid (or *array*) of numbers, symbols, or expressions arranged in rows (horizontal lines) and columns (vertical lines).

$$A^{m \times n} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

Definition: Symmetric matrices

Let a matrix A be *symmetric*, then

$$a_{i,j} = a_{j,i} \quad \forall i, j$$

This is equal to

$$A = A^T$$

Definition: Binary matrices

Let all entries of matrix A only take binary values 0 and 1

One can think of each entry as a logical value where 1 = TRUE and 0 = FALSE

Definition: Matrix Inverse

Much like a function, some matrices have an inverse matrix.

A square matrix $A^{n \times n}$ is called *invertible* (or *non-singular*) if there exists a matrix A^{-1} such that

$$AA^{-1} = A^{-1}A = I_n$$

where I_n is the $n \times n$ identity matrix.

Given a simple matrix $A \in \mathbb{R}^{2 \times 2}$ where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ one can determine

$$A^{-1} = \frac{1}{ac - bd} \cdot \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

If no such A^{-1} exists, A is called *singular* (or *non-invertible*).